Sao Aphisith Sithisack

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EDUCATION

University of Oklahoma **Norman, OK**
B.S. in Electrical Engineering (Honors) | GPA: 3.88 *Aug 2024 – May 2028*
Honors: Davis UWC Scholar (full tuition, 4 yrs) | Robert Vinton Clapp Scholarship
Relevant Coursework: Digital Design, Digital Signal Processing, Introduction to Electronics, Microcontrollers, Circuits Laboratory, Electromagnetic Fields, Differential Equations, Linear Algebra

EXPERIENCE

Sooner Competitive Robotics **Jan 2026 – Present**
Electrical Systems Engineer, Power Distribution *Norman, OK*
- Design the central power and communications board for an autonomous IGVC 2027 robot: a **4-layer Altium** PCB to distribute 12 V at up to **10 A**, CAN, and E-Stop to **7 board ports**, each with INA220 current sensing.
- Specified 12 V input protection (10 A fuse, TVS diode, P-MOSFET reverse-polarity protection), replacing the team-standard Schottky and cutting calculated loss at 8 A from **3.6 W to 0.7 W**.
- Replaced last year's LDO with an AP63301 synchronous buck on the 12 V to 3.3 V rail to reduce heat, raising estimated efficiency from at most 28% to about 55–70% at the 50–100 mA load (datasheet curve).
- Selected the **STM32C092** microcontroller and designed its circuit, with a MAX3051 CAN transceiver, I^2C to 7 INA220 sensors, SWD programming, USB-C to UART debug, and test points so the team can probe faults during bring-up.

Pulsed Laser Template Engineering (PLATEN) Lab **Jan 2026 – Present**
Undergraduate Researcher, Semiconductor Nanofabrication *Norman, OK*
- Fabricate high-aspect-ratio micro- and nanostructures in a cleanroom using photolithography, reactive ion etching (RIE), and pulsed laser deposition (PLD).
- Deposit multiple functional oxide films by pulsed laser deposition (PLD) and characterize them by XRD to explore IC fabrication beyond costly EUV lithography.
- Optimize anisotropic plasma etch parameters (gas chemistry, RF power, chamber pressure) and mask selectivity to reach vertical sidewalls and controlled feature depth.

Student Learning Center **Aug 2025 – Present**
Peer Learning Assistant | 20 hrs/week *Norman, OK*
- Mentor **250+ students per semester** in Physics I–II, Calculus I–III, Differential Equations, Linear Algebra, and General Chemistry I through structured problem-solving.

PROJECTS

TI MSPM0 USB-C Microcontroller Board **Sep 2026 – Present**
KiCad | 2-layer PCB | write-up *Personal Project*
- Designed the schematic for a USB-C powered **Texas Instruments MSPM0G3507** board: 3.3 V LDO, CH340E USB-to-UART bridge, USBLC6 ESD protection, 8 MHz crystal, and SWD debug.

Discrete Analog Function Generator **Oct 2026 – Present**
LTspice *Personal Project*
- Simulated the oscillator core in LTspice, running **Monte Carlo** on resistor tolerance to justify 1% parts: simulated frequency spread narrowed from about $-12\%/+9\%$ at 5% resistors to about $\pm 2\%$.

LEADERSHIP

United World College Lao PDR National Committee **Jul 2024 – Present**
Co-President & Alumni Coordinator *Remote / Laos*
- Co-lead a **20+ volunteer** committee running Laos' UWC scholarship cycle, managing **150+ applicants** per cycle down to 3–4 placements, and led fundraising exceeding **$2,500**.

IEEE Student Branch at OU **Sep 2026 – Present**
Event Committee Member *Norman, OK*
- Plan event logistics and design circuits for technical events, including a 555-timer sound circuit and 12-board DigiKey BOM.

Association of Southeast Asian Nations at OU **Nov 2024 – May 2026**
Sponsorship & Volunteering Chair *Norman, OK*
- Secured $1,000+ in external funding through outreach to 20+ donors and organized cultural events.

TECHNICAL SKILLS

PCB & Hardware: Altium Designer (4-layer PCB), KiCad (schematic capture), power protection and distribution, USB-C, UART, CAN, I^2C, SWD, LTspice (Monte Carlo)
Fabrication & Lab: Photolithography, anisotropic RIE, pulsed laser deposition, X-ray diffraction (XRD), cleanroom processing, oscilloscope, DMM, function generator, bench supply
Programming: C, C++, Python, MATLAB, LabVIEW, Linux, Git